

Pre-Calculus Concepts

ANSWER KEY



Limits: an introductory numerical approach

- 1 Estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.
Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. As $x \rightarrow 2$, this approaches $2 + 2 = 4$.
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Average rate of change

- 2 Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.
Average rate of change $= \frac{f(4) - f(1)}{4 - 1} = \frac{17 - 2}{3} = \frac{15}{3} = 5$.
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Conic sections: ellipses

- 3 Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).
 $\frac{x^2}{25} + \frac{y^2}{9} = 1$.
- 4 For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.
Since $16 > 9$, the major axis is horizontal, with $a = \sqrt{16} = 4$. Vertices: $(4, 0)$ and $(-4, 0)$.
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Conic sections: hyperbolas

- 5 Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.
Since the vertices are horizontal, the equation has the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with $a = 4$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$.
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Parametric equations

- 6 Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x .
From the first equation, $t = x - 1$. Substitute: $y = (x - 1)^2$.
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Sequences and series: sigma notation

- 7 Evaluate $\sum_{k=1}^5 (2k + 1)$
 $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11 = 35$.

- 8 Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.
This is $\sum_{k=1}^5 4k$. Evaluating: $4(1 + 2 + 3 + 4 + 5) = 4(15) = 60$.
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Binomial theorem

- 9 Use the binomial theorem to expand $(x + 2)^4$.
 $(x + 2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16$.
- 10 Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.
The term with x^3 comes from $\binom{5}{3}x^3(3)^2 = 10x^3(9) = 90x^3$. The coefficient is 90.
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Vectors in the plane

- 11 Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.
 $|\mathbf{u}| = \sqrt{9 + 16} = 5$. Unit vector: $\frac{1}{5}(3, -4) = (0.6, -0.8)$.
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A multi-part conic and function analysis problem

- 12 This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.
- (a) Group and complete the square: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$, giving
 $(x - 2)^2 + (y + 3)^2 = 25$. Center: $(2, -3)$. Radius: 5. (b) Distance from center to $(6, -1)$:
 $\sqrt{(6 - 2)^2 + (-1 - (-3))^2} = \sqrt{16 + 4} = \sqrt{20} \approx 4.47$. Since $4.47 < 5$, the point lies INSIDE the circle.
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Matrix operations

- 13 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.
Subtract corresponding entries: first row $(2 - 0, -1 - 5) = (2, -6)$; second row
 $(3 - (-2), 4 - 1) = (5, 3)$.
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Infinite geometric series

- 14 Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \dots$, using $S = \frac{a_1}{1 - r}$.
First term $a_1 = 8$, ratio $r = \frac{1}{2}$. Sum $= \frac{8}{1 - \frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16$.

- 15 Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series

$$\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \dots$$

$$\text{First term } a_1 = \frac{6}{10}, \text{ ratio } r = \frac{1}{10}. \text{ Sum} = \frac{6/10}{1 - 1/10} = \frac{6/10}{9/10} = \frac{6}{9} = \frac{2}{3}.$$

Piecewise and step functions in applied contexts

- 16 A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.

$$C(w) = 5w \text{ for } 0 < w \leq 10, \text{ and } C(w) = 50 + 3(w - 10) \text{ for } w > 10. \text{ For 15 pounds:}$$

$$C(15) = 50 + 3(5) = 50 + 15 = 65 \text{ dollars.}$$

Finding terms of a sequence defined recursively

- 17 A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4 .

$$a_2 = 2(3) + 1 = 7. a_3 = 2(7) + 1 = 15. a_4 = 2(15) + 1 = 31.$$

A multi-part conic section identification and analysis problem

- 18 This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.

(a) Group terms: $4(x^2 - 4x) + 9(y^2 + 2y) = 11$. Complete the square:

$$4(x^2 - 4x + 4) + 9(y^2 + 2y + 1) = 11 + 16 + 9, \text{ giving } 4(x - 2)^2 + 9(y + 1)^2 = 36. \text{ Divide by 36:}$$

$$\frac{(x - 2)^2}{9} + \frac{(y + 1)^2}{4} = 1. \text{ (b) Since both terms are positive and added (not subtracted) with different denominators, this is an ellipse. (c) Center: } (2, -1). \text{ Semi-axes: } a = \sqrt{9} = 3 \text{ (horizontal), } b = \sqrt{4} = 2 \text{ (vertical).}$$